

# Neelkumar K. Shah

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## PROFESSIONAL SUMMARY

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An engineer with 5 years of experience across blockchain development, full-stack engineering, and cybersecurity. Built and hardened Hyperledger Fabric and Ethereum systems for India's National Blockchain Service initiative under the Ministry of Electronics and Information Technology (MeitY). Developed a full-stack security assessment platform (React, Node.js, Express.js, MongoDB) and an automated database vulnerability scanner. Authored 5 peer-reviewed IEEE publications on blockchain security.

## SKILLS

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**Blockchain:** Hyperledger (Fabric, Sawtooth), Ethereum, Solidity, Truffle, Ganache, IPFS, MetaMask

**Development:** Python, Node.js, Express.js, NestJS, React, MongoDB, HTML/CSS, Bootstrap

**Security:** Vulnerability Assessment & Penetration Testing (Web Applications, APIs), Smart Contract Audits (Ethereum, Hyperledger), Static Source Code Analysis, NoSQL Database Security Audit

## PROFESSIONAL EXPERIENCE

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### School of Applied Sciences, Engineering and Technology - Rashtriya Raksha University

Assistant Professor | Gandhinagar, Gujarat, India | Jun 2026 - Aug 2026

- Delivered a full-day training on blockchain and cryptocurrency fundamentals to 20 NCB officers and technical staff from across India.
- Authored Board of Studies (BoS) documentation establishing the curriculum for a new PG Diploma in Blockchain Technology program.

### C-DAC: Centre for Development of Advanced Computing

Project Engineer | Pune, Maharashtra, India | Oct 2021 - Jun 2026

- Led the MeitY-funded initiative to build the National Blockchain Framework (NBF) for India's National Blockchain Service - full technical build detailed under Projects.
- Co-led a 7-person team building a full-stack Blockchain Security Assessment Portal (React, Node.js, Express, MongoDB) automating evaluations across 10-15 APIs, apps, containers, and databases.

### Huptech Web Private Limited

Jr. UI Designer and Developer | Ahmedabad, Gujarat, India | Jun 2019 - Sep 2019

- Built responsive front-end interfaces using HTML, CSS, JavaScript, jQuery, and Bootstrap.

## PROJECTS

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### National Blockchain Framework (NBF) - National Blockchain Service Initiative, MeitY

C-DAC: Centre for Development of Advanced Computing | Oct 2021 - Jun 2026

- Designed and architected a unified, multi-organization Hyperledger Fabric framework (3 peer nodes, 2 organizations, 1 channel) across 2-3 test environments.
- Built REST APIs for a security audit integrating blockchain services.

- Designed and built NBFDBAudit, a Python-based scanner detecting 18+ vulnerability classes across 4-5 MongoDB/CouchDB environments (1 production).
- Led VAPT across 4-5 blockchain applications and 30-45 APIs, surfacing ~25 vulnerabilities per application on average; partnered with engineering on remediation.
- Authored and presented 5 peer-reviewed papers at IEEE ICBDS and I2CT conferences on blockchain and smart contract security.

### **Development and Deployment of E-waste Management Modules Using Blockchain Technology**

Graduate School of Engineering and Technology, GTU (M.E. Dissertation) | Jun 2020 - Jul 2021

- Designed and built a decentralized Ethereum DApp using Solidity, Truffle, Ganache, and IPFS. With implementation of secure, tamper-resistant ownership tracking across the e-waste lifecycle.
- Integrated a React.js frontend with MetaMask wallet connectivity for on-chain interactions.
- Conducted threat modelling and performance optimization of the decentralized architecture.

## **PUBLICATIONS**

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**A Threat Modeling Approach for Securing Blockchain Framework** - 2025 4th IEEE ICBDS

**Security Assessment Framework and Evaluation for Blockchain Applications (SAFE-Block)** - 2024 3rd IEEE ICBDS

**Efficient Solution for NoSQL Database Security in Blockchain-Based Applications** - 2023 2nd IEEE ICBDS

**Smart Contract Vulnerability Detection Techniques for Hyperledger Fabric** - 2023 8th IEEE I2CT

**Smart Contract - Security Assessment Integrated Framework (SC-SIF) for Hyperledger Fabric** - 2022 7th IEEE I2CT

## **EDUCATION**

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### **Master's Degree in Cybersecurity**

Graduate School of Engineering and Technology, GTU, Ahmedabad, Gujarat, India | Sep 2019 - Jul 2021

### **Bachelor's Degree in Computer Engineering**

Marwadi Education Foundation, Rajkot, Gujarat, India | Aug 2015 - May 2019

## **CERTIFICATION & ACHIEVEMENTS**

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- Certified Blockchain Security Professional - Blockchain Council, USA
- Recognized as Performer of the Month - C-DAC, Pune, Maharashtra, India
- Blockchain Technology for Government Officials - Future Skills PRIME, NIELIT, Kolkata, India